QLoRA Fine-Tuning for Next User Turn Prediction and Multi-Step Dialogue Rollouts

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***Abstract*—Most dialogue research optimizes assistant responses, leaving user-turn prediction underexplored. We investigate quantized low-rank adaptation fine-tuning of four open-source instruction-tuned large language models spanning 1.2B to 7B parameters for generating the next user utterance from multi-turn dialogue context. Models are trained on 6,000 samples from two complementary corpora, one open-domain and one task-oriented, and evaluated against pretrained models, GPT-4o-mini prompt baselines using automated metrics, short multi-step rollouts, and a blinded human study of 369 samples. Fine-tuning consistently improves fluency and semantic quality across three of four model families, with Llama-3B achieving the highest single-turn performance. Reference-assisted rollouts show stable three-step behavior for the strongest fine-tuned models. Human evaluation confirms a 0.34 average improvement on a 5-point scale, with task-oriented dialogue benefiting more (0.45) than open-domain conversation (0.21). A systematic ablation identifies the optimal LoRA configuration. Our results show that fine-tuning enables practical and reproducible user-turn prediction, and that short rollout evaluation is necessary to separate single-turn gains from longer-horizon stability..**

***Index Terms*—Large Language Models, Low-Rank Adaptation, QLoRA, Parameter-Efficient Fine-Tuning (PEFT), Dialogue Systems, User-turn Prediction**

I. Introduction

Conversational AI has progressed from rule-based systems to end-to-end generative models built on the transformer architecture [1], [2]. The vast majority of this research focuses on improving the *assistant's* response quality, while user behavior is typically modeled with hand-crafted templates or reinforcement-learning agents [3]. This imbalance limits the realism of dialogue evaluation and the ability to stress-test systems against diverse user behavior.

Predicting the next user turn is inherently more challenging than generating assistant responses: user utterances are less constrained, span a wider range of intents, and depend heavily on conversational context. Naous et al. [4] recently demonstrated that assistant-trained LLMs make poor user simulators, with the counterintuitive finding that stronger assistants produce worse user simulations. Accurate generative user modeling would enable realistic dialogue simulation, continuous system evaluation, and stress testing without requiring human participants.

In this paper, we investigate QLoRA fine-tuning [5] of four open-source instruction-tuned LLMs (1.2B–7B parameters) for next user turn prediction. We train on 6,000 samples drawn equally from an open-domain corpus (WildChat [6]) and a task-oriented corpus (Schema-Guided Dialog [7]), and compare against pretrained base models. We further evaluate short multi-step rollouts to test whether one-step improvements remain stable when generated user turns are fed back into the dialogue context.

To structure our investigation, we pose the following research question:

"Can QLoRA-based fine-tuning improve user-turn prediction quality across open-source instruction-tuned large language models under low-data constraints, and do these improvements differ between open-domain and task-oriented dialogues, and do they remain stable in short multi-step rollouts"

Our contributions are as follows:

1) A multi-model QLoRA study for generative user-turn prediction, evaluating four architectures across open-domain and task-oriented dialogue.
2) A systematic two-stage ablation of LoRA hyperparameters comprising 56 experiments.
3) A combined evaluation using three automated metrics, multi-step rollouts, and a blinded human study of 369 samples.
4) Evidence that fine-tuning yields substantially larger gains on task-oriented dialogue than on open-domain conversation.

II. Related Work

A. User Simulation and Turn Prediction

User simulators have traditionally relied on rule-based or agenda-based architectures [3]. Neural approaches extended this: Kim and Lipani [8] jointly predict user actions and utterances in goal-oriented settings; Lin et al. [9] introduced

GenTUS for task-oriented user simulation; Yoon et al. [10] evaluated LLMs as user simulators for conversational recommendation, finding discrepancies with real behavior. Most closely related, Naous et al. [4] introduced *User Language Models*, showing that assistant-trained LLMs make poor user simulators. Their approach uses full fine-tuning on over one million samples with intent conditioning, requiring 227 GPU-hours on four A6000s. We differ by using QLoRA [5] on instruction-tuned models with only 6,000 samples and no intent conditioning, enabling cross-domain evaluation with four model families (1.2B–7B) on a single GPU.

B. Parameter-Efficient Fine-Tuning

LoRA [11] freezes pretrained weights and injects trainable low-rank matrices into transformer layers, reducing trainable parameters by over 99%. QLoRA [5] extends this with 4-bit quantization and paged optimizers, enabling fine-tuning of large models on a single GPU. These methods allow controlled study of behavioral adaptation—shifting from assistant-style to user-style generation—without full retraining, making experiments reproducible on academic hardware.

C. Evaluation of Generated Dialogue

Liu et al. [12] showed that word-overlap metrics correlate weakly with human judgments for dialogue. BERTScore [13] uses contextual embeddings to capture semantic similarity beyond surface matching. BLEURT [14] is a learned metric robust to distribution shifts. Perplexity measures token-level fluency. We combine all three with a blinded human evaluation on a 1–5 Likert scale.

D. Gap Analysis

Table I summarizes the coverage of key research dimensions in prior work. While individual dimensions are addressed by several studies, none combine all five. Our work fills this gap by applying QLoRA to four architectures for generative user-turn prediction, evaluated across open-domain and task-oriented dialogue with automated metrics and a blinded human study.

TABLE I: Coverage of Research Dimensions in Related Work

Citation	User-turn generation	PEFT	Multi-model	Cross-domain	Human evaluation
Shi et al. [3]	✓	✗	✗	✗	✓
Kim & Lipani [8]	✓	✗	✓	✓	✗
Lin et al. [9]	✓	✗	✗	✗	✓
Yoon et al. [10]	✓	✗	✓	✗	✗
Naous et al. [4]	✓	✗	✗	✗	✓
Hu et al. [11]	✗	✓	✓	✗	✗
Dettmers et al. [5]	✗	✓	✓	✗	✓
This paper	✓	✓	✓	✓	✓

III. Methodology

A. Task Definition

Our pipeline (Fig. 1) takes a multi-turn conversation history $C = (u_1, a_1, u_2, a_2, \ldots, a_{t-1})$ and generates the next user utterance $\hat{u}_t$, where u_i and a_i are user and assistant turns, respectively. The model receives C formatted as a chat template and produces $\hat{u}_t$ autoregressively. During training, only target user-turn tokens are supervised, while all context tokens are masked (label = -100), focusing the model on user-turn generation. Evaluation uses both one-step prediction and short multi-step rollouts. In the rollout setting, we seed each dialogue with an initial user–assistant exchange, generate the next user turn, append it to the history, and continue for up to three user turns.

B. Datasets

We use two complementary datasets to evaluate cross-domain generalization. **WildChat-1M** [6] contains one million real user–ChatGPT conversations spanning diverse topics and languages; we filter for English, multi-turn (≥3 turns) dialogues ending with an assistant turn, yielding context–target pairs for single-turn prediction and full trajectories for rollout evaluation. **Schema-Guided Dialog** (SGD) [7] provides over 16k task-oriented dialogues across 16 service domains with structured intents. To explore PEFT under limited data, we sample 3,200 pairs per dataset (6,400 total; Table II), which is three orders of magnitude less than the 1M+ samples used by Naous et al. [4]. This setup evaluates whether meaningful adaptation can be achieved on small, balanced datasets, making the approach feasible for research groups with constrained compute resources. Pairs are interleaved to maintain domain balance, and a fixed seed (42) ensures reproducibility.

The full pipeline and code are publicly available.[1]

TABLE II: Dataset Statistics

Dataset	Train	Eval	Total
WildChat-1M	3,000	200	3,200
Schema-Guided Dialog	3,000	200	3,200
Total	6,000	400	6,400

[1] https://github.com/Prof-it/user_turn_lora

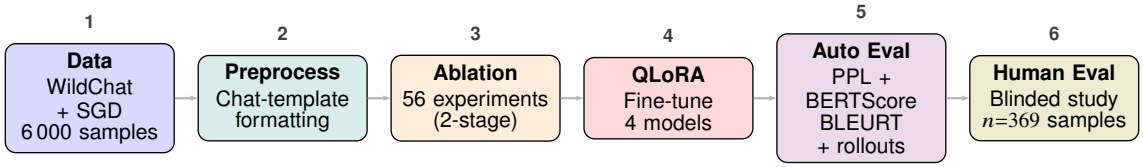

Fig. 1: End-to-end research pipeline from data curation through automated and human evaluation.

C. Models

We evaluate four open-source instruction-tuned LLMs spanning 1.2B to 7B parameters (Table III), selected to test whether LoRA-based user-turn adaptation generalizes across architectures and model scales. All models use chat-template formatting with system, user, and assistant roles. As a prompt-only baseline, we include GPT-4o-mini in zero-shot and few-shot (3 exemplars) configurations, queried via API at temperature 0.4 with a maximum of 256 tokens.

TABLE III: Models Evaluated

Model	Parameters
Qwen2.5-3B-Instruct	3B
Llama-3.2-3B-Instruct	3B
OLMo-3-7B-Instruct	7B
LFM2.5-1.2B-Instruct	1.2B
GPT-4o-mini (prompt baseline)	–

D. Training Setup

All models are trained with QLoRA [5]: base weights quantized to 4-bit NormalFloat (bfloat16 compute), paged AdamW 8-bit optimizer, cosine schedule, gradient checkpointing, 4,096-token context, and effective batch size 64. All experiments run on a single NVIDIA A100-40GB GPU with identical configuration across all four models.

E. LoRA Configuration

The LoRA hyperparameters in Table IV were determined through a systematic two-stage ablation study on the 1.2B model (Section IV-D) and then applied to the other model families for the main comparison. LoRA adapters are applied to all linear projection layers in each transformer block (query, key, value, output, gate, up, and down projections). Because larger models may require different scaling factors, we additionally run a targeted OLMo-7B sweep over $\alpha \in \{16, 32, 64\}$ and learning rate $\in \{1\times10^{-4}, 2\times10^{-4}\}$. Notably, a rank of just 8 performs comparably to 64, while the scaling factor $\alpha = 64$ is the most impactful parameter, suggesting that user-turn adaptation resides on a very low-rank subspace.

TABLE IV: Best configuration identified on the 1.2B model and applied to other models (from Ablation)

Parameter	Value
LoRA rank (r)	8
LoRA alpha (α)	64
LoRA dropout	0.0
Learning rate	2×10^{-4}
Epochs	3
Warmup ratio	0.1
Weight decay	0.0
Target modules	q, k, v, o, gate, up, down

IV. Experiments

A. Evaluation Metrics

We report three automated metrics. **Perplexity** (PPL) measures token-level fluency; lower values indicate the model assigns higher probability to ground-truth tokens. **BERTScore** F1 [13] computes contextual embedding similarity, capturing semantic overlap beyond surface-level word matching. **BLEURT** [14] is a learned metric trained on human judgments that is robust to distribution shifts. For rollouts, we report macro BERTScore and BLEURT over generated user turns, step-wise degradation from the first to final rollout step, and collapse rate, defined as the fraction of dialogues ending early due to empty or repeated user generations or empty assistant generations. For human evaluation, a single rater assessed 369 blinded samples across four conditions (baseline, fine-tuned, GPT-4o-mini zero-shot, GPT-4o-mini few-shot) on relevance, coherence, and naturalness using a 1–5 Likert scale with randomized labels (A–D).

B. Baselines

Each fine-tuned model is compared against three baselines: (1) the *base model* without fine-tuning, measuring what the pretrained instruction-tuned model already captures; (2) *GPT-4o-mini zero-shot*, where the model receives a system instruction and conversation context with no examples; and (3) *GPT-4o-mini few-shot*, which additionally includes three fixed exemplar context–user-turn pairs drawn from the training set. The prompt baselines test whether a larger proprietary model

can solve the task without any training data. They also probe a fundamental limitation of API-based models: since these models are instruction-tuned to respond as assistants, they must be explicitly prompted to adopt a user role, which may conflict with their training objective.

C. Rollout Protocol

To address error accumulation, we evaluate short rollouts on reconstructed WildChat and SGD trajectories. We use a reference-assisted protocol as the main rollout setting: the simulator generates a user turn, the generated turn is appended to the history, and the next gold assistant turn from the dataset is appended before the following user prediction. This isolates user-simulator drift without confounding results with assistant-model errors. We also run a smaller closed-loop setting with a fixed Qwen2.5-3B-Instruct assistant, where both user and assistant turns are generated. All rollouts use one seed user–assistant pair and a maximum of three generated user turns.

D. Ablation Study

We conducted a two-stage ablation on LiquidAI/LFM2.5-1.2B-Instruct, the smallest model in our study, to enable rapid iteration. Stage 1 searched over LoRA architecture parameters (rank $r \in \{8, 16, 32, 64\}$, $\alpha \in \{16, 32, 64\}$, dropout $\in \{0.0, 0.05\}$; 24 experiments). Stage 2 fixed the best LoRA configuration and searched over training hyperparameters (learning rate, epochs, warmup ratio, weight decay; 32 experiments). The full ablation comprised 56 experiments completed in approximately 4 hours on a single A100 GPU. The resulting optimal configuration (Table IV) was applied to all four model families. Because OLMo-7B showed degraded semantic metrics under this transferred configuration, we additionally ran a targeted six-run OLMo-7B sweep over $\alpha \in \{16, 32, 64\}$ and learning rate $\in \{1\times10^{-4}, 2\times10^{-4}\}$. Larger models may require different scaling factors, though.

E. Temperature Sweep

We evaluated five temperature settings (0.3, 0.4, 0.5, 0.6, 0.7) for both baseline and fine-tuned variants of each model. Lower temperatures generally yield higher BERTScore and BLEURT, as more deterministic generation produces outputs closer to the single ground-truth reference.

V. Results

A. Automated Metrics

Table V compares baseline and fine-tuned models on all three automated metrics. Fine-tuning yields consistent perplexity reductions across all four models, with the largest gains for LiquidAI-1.2B (409.6→9.4, a 97.7% reduction) and Llama-3B (100.6→11.2, 88.9%). BERTScore and BLEURT improve for three of four models, with Llama-3B achieving the highest overall scores (0.854 BERTScore, 0.393 BLEURT). The exception is OLMo-7B, where fine-tuning improves perplexity (11.1→9.0) but decreases BERTScore (0.813→0.798) and BLEURT (0.285→0.229), an anomaly we discuss in Section VI (see also Fig. 2).

TABLE V: Automated Evaluation Results (Baseline vs Fine-tuned)

	Baseline			Fine-tuned		
Model	BERT	BLEURT	PPL	BERT	BLEURT	PPL
Qwen-3B	0.840	0.374	62.9	0.846	0.384	**10.4**
Llama-3B	0.828	0.338	100.6	**0.854**	**0.393**	11.2
OLMo-7B	0.813	0.285	11.1	0.798	0.229	9.0
LiquidAI-1.2B	0.756	0.323	409.6	0.837	0.373	9.4

The targeted OLMo-7B follow-up sweep confirms that the transferred 1.2B configuration is not universally optimal. The best OLMo setting in this six-run sweep uses α=64 and learning rate 2×10^{-4}, reaching 0.811 BERTScore, 0.245 BLEURT, and 8.9 perplexity. This improves over the original OLMo fine-tuned run on semantic metrics but remains below the OLMo base model's BLEURT, so we treat OLMo as evidence of scale-specific tuning sensitivity rather than as a successful adaptation case.

TABLE VI: Prompt Baseline Results

Model	Zero-shot BERT	Zero-shot BLEURT	Few-shot BERT	Few-shot BL
LiquidAI-1.2B	0.833	0.369	0.812	0.385
Qwen-3B	0.832	0.367	0.826	0.363
Llama-3B	0.833	0.336	0.830	0.340
OLMo-7B	0.840	0.391	0.828	0.380
GPT-4o-mini	0.841	0.393	**0.847**	**0.401**

Table VI shows the prompt baseline results. GPT-4o-mini achieves competitive BERTScore (0.841 zero-shot, 0.847 few-shot) and BLEURT (0.393, 0.401), matching or slightly exceeding the fine-tuned Qwen-3B (0.846, 0.384). Few-shot prompting does not consistently improve the open-source base models: LiquidAI improves on BLEURT but loses BERTScore, while Qwen and OLMo decrease. This indicates that the gains in Table V are not explained by in-context learning alone. Perplexity is not reported for prompt baselines as the API does not expose token-level log-probabilities under the same formulation.

B. Temperature Analysis

Table VII reports the optimal temperature for each fine-tuned model. Three of four models perform best at temperature 0.3, confirming that more deterministic decoding better matches the single ground-truth reference. OLMo-7B is the exception, preferring 0.6, consistent with its anomalous behavior on semantic metrics, where higher diversity may partially compensate for reduced alignment.

TABLE VII: Optimal Temperature per Model (Fine-tuned)

Model	Best Temp	BERTScore
Qwen-3B	0.3	0.859
Llama-3B	0.3	0.864
OLMo-7B	0.6	0.821
LiquidAI-1.2B	0.3	0.857

C. Multi-Step Rollouts

Table VIII reports the full-dataset rollout evaluation over balanced WildChat and SGD trajectories. The reference-assisted setting measures user-simulator error accumulation while holding assistant behavior fixed to the dataset trajectory. Degradation is computed as first-step minus final-step user BERTScore; lower values indicate more stable multi-step behavior.

TABLE VIII: Reference-Assisted Multi-Step Rollout Results on WildChat+SGD

Model	Condition	User BERT	User BLEURT	Degradation	Collapse
LiquidAI-1.2B	Base	0.755	0.298	0.064	0.16
LiquidAI-1.2B	Fine-tuned	0.855	0.366	0.033	0.02
Qwen-3B	Base	0.839	0.352	0.016	0.00
Qwen-3B	Fine-tuned	0.858	0.347	0.007	0.02
Llama-3B	Base	0.823	0.357	0.008	0.00
Llama-3B	Fine-tuned	**0.864**	**0.377**	0.013	0.00
OLMo-7B	Base	0.819	0.302	0.005	0.00
OLMo-7B	Fine-tuned	0.799	0.218	**0.004**	0.10

We additionally run closed-loop rollouts with a fixed Qwen2.5-3B-Instruct assistant to verify that reference-assisted stability is not solely an artifact of using gold assistant turns. These closed-loop results are used as a robustness check rather than the main comparison, because assistant errors and user-simulator errors are coupled in that setting.

D. Human Evaluation

Table IX presents the blinded human evaluation results (mean ± std over 369 paired samples). The human evaluation should be interpreted as indicative rather than definitive due to the use of a single annotator. Fine-tuning improves the average rating from 4.37 ± 1.19 to 4.71 ± 0.75 (+0.34), with the largest gain on naturalness (+0.37). A paired Wilcoxon signed-rank test confirms this improvement is highly significant ($p < 10^{-5}$). GPT-4o-mini scores highest overall (4.82 ± 0.62), though the gap to the fine-tuned model is modest; the fine-tuned model significantly differs from the few-shot baseline ($p < 0.01$). The domain breakdown reveals that task-oriented dialogue (SGD) benefits substantially from fine-tuning (+0.45, $p < 10^{-8}$), whereas the observed gain on open-domain conversation (WildChat, +0.21) was not statistically significant ($p = 0.12$, n.s.), suggesting that structured intent patterns are more readily learned from limited data.

VI. Discussion

QLoRA fine-tuning consistently improves user-turn prediction across three of four model families. BERTScore and BLEURT gains confirm semantically more appropriate outputs, not merely more probable token sequences, and the blinded human evaluation reinforces this (+0.34 on a 5-point scale).

A notable domain asymmetry emerges: fine-tuning yields a substantial gain on task-oriented dialogue (SGD, +0.45, $p < 10^{-8}$), whereas the observed gain for open-domain conversation (WildChat, +0.21) was not statistically significant (p = 0.12). This suggests that structured intent patterns are more readily learned from limited data, while open-domain dialogues may require more data or different adaptation strategies to achieve meaningful improvement.

The rollout protocol directly tests whether one-step prediction gains remain usable after generated user turns enter the context. The reference-assisted setting is intentionally conservative: it isolates user-simulator drift while preventing assistant errors from dominating the measurement. The closed-loop setting with a fixed assistant provides a secondary robustness check, but it is not used as the primary comparison because both agents can introduce errors.

OLMo-7B is the only model in which fine-tuning degrades semantic metrics (BERTScore 0.813→0.798) despite improving perplexity (11.1→9.0). Since the original ablation was conducted on the 1.2B model, the learning rate (2×10^{-4}) and α=64 may over-adapt the 7B model's larger parameter space, causing it to memorize surface patterns (lower perplexity) while losing semantic generalization. The targeted OLMo-7B sweep improves the fine-tuned BLEURT score to 0.245 but does not close the gap to the base model, confirming that hyperparameter transfer across model scales must be treated cautiously. A per-model ablation would likely resolve this.

The GPT-4o-mini prompt baselines score highest in human evaluation (4.80–4.82), yet the fine-tuned Llama-3B surpasses

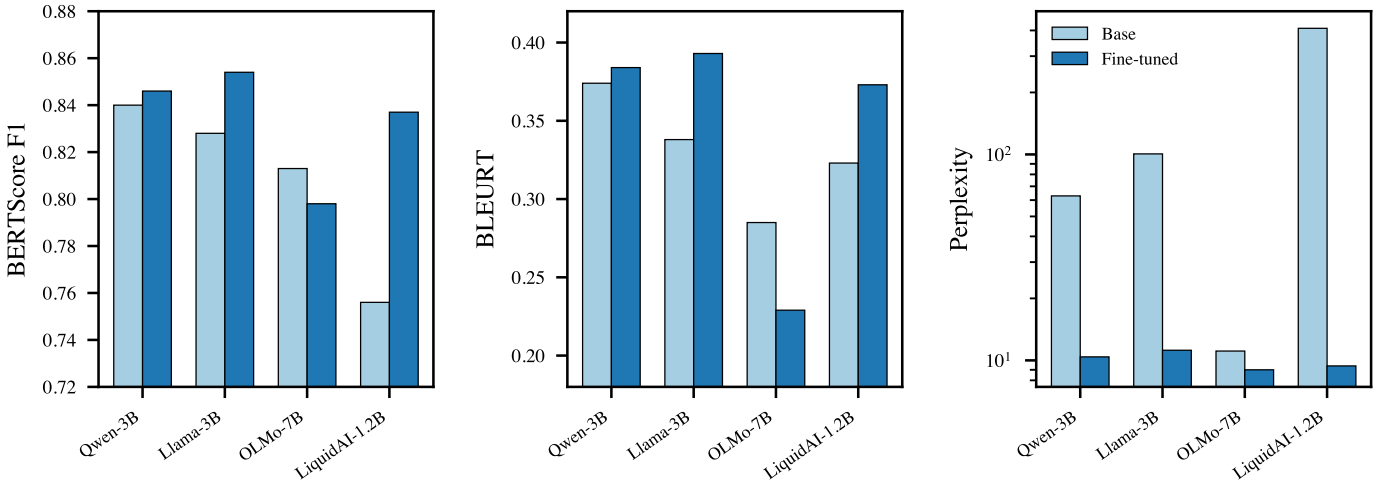

Fig. 2: Base vs. fine-tuned comparison across all four models on BERTScore F1, BLEURT, and perplexity (log scale). Fine-tuning improves all metrics for three of four models. OLMo-7B is the exception: perplexity decreases but BERTScore and BLEURT degrade after fine-tuning.

TABLE IX: Human Evaluation (mean ± std, n=369, 1–5 Likert)

Condition	**Rel.**	**Coh.**	**Nat.**	**Avg.**
Baseline (Qwen-3B)	4.34±1.23	4.34±1.25	4.44±1.29	4.37±1.19
Fine-tuned (Qwen-3B)	4.64±0.88	4.68±0.85	4.81±0.75	**4.71**±0.75
Zero-shot (GPT-4o-mini)	4.78±0.69	4.82±0.67	4.85±0.66	4.82±0.62
Few-shot (GPT-4o-mini)	4.77±0.75	4.80±0.72	4.83±0.76	4.80±0.72
Domain breakdown (Avg.):				
SGD only (n=200)	Base: 4.50±1.06		FT: **4.95**±0.22 (Δ=+0.45***)	
WildChat only (n=169)	Base: 4.22±1.31		FT: 4.43±1.01 (Δ=+0.21$^{\text{n.s.}}$)	

Wilcoxon signed-rank: $^{***}p<0.001$; $^{\text{n.s.}}$ not significant (p=0.12).

GPT-4o-mini on automated metrics (0.854 BERTScore, 0.393 BLEURT), demonstrating that PEFT of small open models can exceed large proprietary alternatives.

The ablation reveals that α is the most impactful LoRA hyperparameter, while rank has minimal effect (r=8 $\approx$ r=64), consistent with Hu et al. [11]. The temperature sweep confirms that 0.3 optimizes metrics for most models; OLMo diverges at 0.6.

Implications. Our results suggest that user-turn adaptation lies on a low-rank subspace that PEFT can capture with minimal data. This makes realistic user simulation accessible to research groups without large compute budgets, enabling continuous dialogue system evaluation, data augmentation for underrepresented domains, and stress-testing of deployed assistants—all without human participants.

Limitations. The human evaluation relies on a single rater. The main ablation was conducted on the smallest model, and the OLMo follow-up shows configurations may not transfer across scales. All experiments are English-only, and evaluation uses a single ground-truth user-turn reference. The rollout evaluation is deliberately short and mostly reference-assisted, si it measures local error accumulation rather than long-horizon task success in fully interactive simulations. Future work should assess performance in closed-loop settings where errors may accumulate across turns. Additionally, the study focuses exclusively on QLoRA; alternative parameter-efficient methods or reinforcement-based strategies may reveal different trade-offs between data efficiency and conversational realism.

VII. Conclusion

We investigated QLoRA fine-tuning for next user turn prediction and short multi-step rollout stability across four open-source LLMs (1.2B–7B). Training on 6,000 samples yields

consistent improvements for three of four models, with Llama-3B achieving the best single-turn scores (0.854 BERTScore, 0.393 BLEURT). A blinded human evaluation (n=369) confirms a +0.34 average improvement overall, driven primarily by task-oriented dialogues (+0.45, $p < 10^{-8}$), while the observed gain for open-domain conversation (+0.21) was not statistically significant ($p = 0.12$).

These results demonstrate that realistic user-turn simulation can be achieved with limited data and modest computational resources (single GPU), without relying on large proprietary models. This makes it feasible for smaller research groups and enterprises to conduct cost-effective evaluation, stress testing, and iterative development of conversational systems.

While OLMo-7B showed degraded semantic metrics, the targeted follow-up sweep confirms that careful per-model hyperparameter tuning is necessary, highlighting a limitation in scaling QLoRA uniformly across architectures. Evaluation was conducted on single-turn predictions in English, with a single human rater, and the rollout study is limited to short trajectories, which constrains claims about long-horizon robustness.

Future work should explore longer closed-loop rollouts, multilingual data, domain-specific task-success metrics, and alternative parameter-efficient or reinforcement-learning methods to further enhance user-turn modeling and cross-domain generalization.

Overall, this study shows that low-rank fine-tuning provides a practical and reproducible path to improve user-turn prediction, bridging a key gap in conversational AI research.

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